

# An explainable algorithm for credit risk prediction based on contrast patterns

Erwis Melchor Pérez<sup>1</sup>[0000–1111–2222–3333], José Francisco  
Martínez-Trinidad<sup>1</sup>[1111–2222–3333–4444], and Jesús Ariel Carrasco  
Ochoa<sup>1</sup>[2222–3333–4444–5555]

Instituto Nacional de Astrofísica, Óptica y Electrónica  
{emelchor, fmartine, ariel}@inaoep.mx

**Abstract.** Resumen

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## 1 Introduction

Credit risk prediction involves determining whether an applicant will default on a future financial obligation [1], and is therefore a fundamental task in the financial sector. Credit risk prediction is commonly considered as a supervised classification task; the goal is to assign each applicant to a class (e.g., default or non-default), rather than estimating a continuous predictive risk value. Credit risk prediction directly impacts the stability of banks and financial institutions, where accurate and explainable predictions are necessary to ensure regulatory compliance [2, 3]. Increasingly, financial institutions have implemented evaluations and management processes that employ pattern recognition and machine learning algorithms for credit risk prediction [4, 5]. As these predictors become more widely adopted, improving default prediction enables more accurate credit risk prediction and reduces losses [6–8].

Recent research in credit risk prediction has prioritized improving the performance of credit risk prediction algorithms based on machine learning and pattern recognition through the use of oversampling or undersampling methods, thereby reducing algorithmic bias towards the majority class [9–13]; the use of feature selection methods for identifying the most relevant features of a dataset to improve the performance of prediction algorithms [1, 14–16]; and the use of explainable AI techniques [10, 17–19] that allow users to comprehend and trust the predictions provided by credit risk prediction algorithms, which makes the predictions of the algorithm to comply with regulations.

Machine learning algorithms have demonstrated good predictive performance, and decision trees are among the most effective for credit risk prediction [4, 20–25]. However, the need for explanations remains a significant limitation, particularly in the financial sector, as financial institutions and regulators require

justification for their decisions [26, 27]. Several authors have addressed the lack of explainability in credit risk prediction [17, 28, 29].

Explainability techniques, also known as XAI (eXplainable Artificial Intelligence), aim to make machine learning prediction processes transparent and understandable to humans. These techniques are important in areas such as credit risk prediction, where justifying and explaining the predictions is essential in enabling financial institutions to make more informed decisions. Several recent studies have applied SHAP [30] and LIME [31] to credit risk prediction to generate explanations for individual predictions [32–34]. SHAP assigns to each feature a value that represents how much it contributed to the prediction. These values help explain the role each feature played in the prediction. LIME builds a simple interpretable predictor that approximates the behaviour of the original predictor by modifying a specific instance and observing the changes in the prediction. The result is an explanation that highlights the features contributing most to the prediction, showing how each affects the prediction in a positive or negative direction. However, these techniques focus on individual features and often overlook how several features jointly contribute to the prediction.

In credit risk prediction, a predictor’s output depends on multiple features rather than individual features. Therefore, explainable predictors should consider multiple features to explain the prediction [18, 35]. In response to this need, this paper proposes a credit risk prediction algorithm that generates explanations based on contrast patterns, which are conjunctions of conditions of the form *feature#value* that are frequent in a class but rarely appear in the opposite class. Since contrast patterns can be extracted from decision-tree-based, we propose using decision-tree-based predictors to predict credit risk, which have been reported to perform competitively compared to the most successful approaches. In this way, the proposed algorithm leverages the affinity and synergy between decision-tree-based predictors, and contrast patterns extracted from decision treed.

Unlike other explainable algorithms that merely highlight the importance of individual features, our method explains predictions based on contrast patterns present in the training data and shared with the new applicant. Experiments conducted on both public and private financial datasets show that the proposed algorithm achieves competitive predictive accuracy while offering instance-level explanations based on the contrast patterns shared with other clients in the class, unlike the current state-of-the-art, which explains based on the single features contributing most to the prediction.

The remainder of this paper is organized as follows. Section 2 presents the related work. Section 3 describes the proposed algorithm. Section 4 presents our experimental results and a discussion about them. Finally, Section 5 presents our conclusions and future work directions.

## 2 Related works

Credit risk prediction has been extensively studied in the literature using a wide range of pattern recognition and machine learning algorithms [3, 7, 8, 10, 14, 15, 28, 29, 33, 36–48]. In this section, we review the most recent works on credit risk prediction, with a focus on approaches that either prioritize achieving the highest accurate credit risk prediction or incorporate explainability for credit risk prediction. The works considered in this review are those that utilize publicly available datasets, commonly used in the literature and accessible online, which allows a comparison against our proposal. Accordingly, this section is divided into two parts: works that focus solely on accuracy without offering explanations, and works that integrate explainability into their predictors.

### 2.1 Predictors for Credit Risk Prediction

Recent works on credit risk prediction have focused on developing predictive algorithms to improve performance metrics, such as accuracy, area under the curve (AUC), Type II error, or F1-score.

In [38], the authors proposed a stacked ensemble for credit risk prediction that combines Random Forests (RF), Logistic Regression (LR), and Convolutional Neural Networks (CNN). The results of these predictors are used as input features for a Multilayer Perceptron (MLP), which learns to combine them to produce the prediction. The authors used SMOTEENN [49] to address class imbalance, and the stacked ensemble is compared against CNN, LR, MLP, and RF. After applying SMOTE-ENN to address class imbalance, the stacked ensemble consistently outperformed the individual predictors, achieving accuracies near 96%.

A comparative study of different predictors for credit risk prediction is presented in [46], including LR, RF, and Artificial Neural Networks (ANN). In this study, SMOTE [50] was employed to balance the training dataset, leading to significant improvements in the performance metrics of all predictors. Among them, RF achieved the best results, with an accuracy of 98%.

In [47], Deep Convolutional Neural Networks (DCNN) and Support Vector Regression (SVR) were compared for credit risk prediction. The application of SMOTE contributed to improved predictive performance in both predictors. Using SMOTE to balance the dataset, SVR achieved an accuracy of 98% and F1-score of 87%.

In [51], the authors present the Level-Wise Feature-Guided Cascading Ensembles (LFGCE) model, which integrates hierarchical feature selection with a cascading ensemble learning scheme. To evaluate its performance, several predictors were employed, including RF, Support Vector Machines (SVM), LR, Decision Trees (DT), ANN, Linear Discriminant Analysis (LDA), and XGBoost.

Among these, XGBoost achieved the best results in both feature selection and prediction, with an AUC of 94%. However, it is important to note that the study does not address class imbalance, as no data balancing methods were applied.

Most of the reviewed studies emphasise improving predictive performance through class balancing methods to mitigate data imbalance. Several predictors yield good results in credit risk prediction. However, the reported performance metrics (such as accuracy, AUC, and F1-score) vary significantly on the chosen predictors and balancing methods. Overall, the authors consistently point out that addressing class imbalance is crucial for achieving reliable predictions, with many studies reporting high accuracy values, often exceeding 90%.

## 2.2 Explainable Predictors for Credit Risk Prediction

Incorporating explainability into credit risk prediction has been a growing area of study. These studies aim to achieve robust predictive performance while prioritising the explainability of predictions.

A study by [43] addresses credit risk prediction by comparing the predictive performance of XGBoost, RF, Least Squares SVM (LSSVM), and a CNN. After applying SMOTE to balance the dataset, XGBoost achieved the best predictive accuracy, reaching 92%. To enhance predictor explainability, SHAP analysis was applied to quantify the individual contributions of each feature. From the predictors evaluated, XGBoost performed best in terms of predictive accuracy and also provided explanations through SHAP values.

In [33], the authors present an ensemble framework that combines RF and XGBoost using probability-weighted averaging. The framework was evaluated using eight predictors, including LR, DT, Gradient Boosting (GB), LightGBM (LGBM), MLP, Naive Bayes (NB), and XGBoost. Their proposed ensemble significantly outperformed all others, achieving an AUC of 97%. SHAP was employed to explain the predictions, which allowed them to determine the contribution of each feature.

In [29], XGBoost is used as a predictor, combined with SHAP, LIME, and K-means to enhance explainability, achieving an AUC of 78%. SHAP and LIME are used to generate feature-contribution vectors for each training instance, which are grouped using K-means. For explaining the prediction of a new instance, its feature-contribution vector is calculated and assigned to the nearest cluster, and the prediction explanation is provided as the cluster’s average vector.

In [3], the authors present a comparative analysis of the performance of boosting algorithms, including LightGBM, XGBoost, CatBoost, and AdaBoost. Additionally, SMOTE-Tomek [52] is employed to balance the dataset. The results indicate that LightGBM achieves the best performance with 95% accuracy and 98% AUC, outperforming the other boosting methods. SHAP is applied to

quantify the contribution of each feature to the prediction.

In [28], a multilayer dynamic graph neural network (DYMGN), proposed for credit risk prediction. The authors compare their approach against several baselines: LR, XGBoost, and DNN. Unlike many related studies, no resampling or class balancing methods are applied. Experimental results show that the DYMGN achieves the best results with 81% accuracy and 83% F1-score. For explainability, SHAP is used to identify the features that contribute most to the predictions.

In [48], the authors propose an explainable two-stage technique to explain the results of credit risk prediction. In the first stage, SHAP is applied to analyse the importance of features according to their positive or negative impact on the prediction. In the second stage, they use a genetic algorithm to generate counterfactual explanations by perturbing the features according to this categorisation, thereby identifying the minimum necessary changes to modify the prediction. This technique was evaluated using multiple predictors, including SVM, XGBoost, MLP, and LSTM, with an accuracy of 90% for the employed predictors. The results demonstrate that the features modified most frequently closely align with the features identified as most relevant by SHAP.

In [53], the authors compare the performance of several Deep Learning (DL) architectures for credit risk prediction, including Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM) networks, Gated Recurrent Units (GRU), and Graph Neural Networks (GNN). To address class imbalance, they apply SMOTEENN. The experimental results show that GRU outperformed all other predictors, achieving an accuracy of 91%. To explain the prediction, SHAP is used to quantify the importance of features and their contributions to the predictions.

In [7], the authors propose an approach for credit risk prediction on imbalanced datasets by integrating the TabTransformer model with a weighted loss function. This approach helps mitigate the bias toward the majority class, improving predictive performance on minority classes, achieving 91% accuracy and an AUC of 93%. SHAP is employed to evaluate feature importance and identify those features that contribute most to the predictions.

In [54], the authors propose a robust framework for credit risk prediction on imbalanced datasets by introducing the K-SMOTEENN method, which integrates oversampling and noise-filtering methods with a stacking ensemble. The ensemble combines Random Forest, XGBoost, and LightGBM, resulting in a predictive performance of 92% F1-score, and 96% AUC. SHAP is used to analyse and explain the contributions of individual features to the prediction.

As discussed in the above works, credit risk prediction has been addressed through a wide range of predictors, balancing methods, feature selection methods, and evaluated through multiple performance metrics. Nevertheless, accurately detecting defaults remains a significant challenge, as no single method has consistently outperformed others across different scenarios. Moreover, the most commonly used explainability approaches, such as SHAP and LIME, while helpful in assessing the individual contribution of features, fail to capture feature value combinations that characterise different types of applicants. This limitation is particularly relevant, since financial institutions increasingly demand not only high predictive accuracy but also clear and case-specific justifications, both to meet regulatory requirements and to improve communication with applicants.

### 3 Proposed explainable algorithm

As discussed in the previous section, recent research has increasingly focused on explaining credit risk predictions. Techniques such as SHAP and LIME, commonly used for explaining credit risk predictions, assign importance scores to individual features; however, they do not capture the joint effect of multiple features in a prediction, which is particularly relevant in credit risk prediction. This limitation highlights the need for approaches that go beyond individual feature contributions and consider multiple features when explaining credit risk predictions.

The idea behind the proposed Pattern-based Credit Risk Prediction and Explanation (PbCRPE) algorithm is to enable both credit risk prediction and its explanation to rely on the same approach. Unlike traditional explainers for credit risk prediction, which apply a predictor that follows an approach different from the one used for explaining predictions (e.g., SHAP or LIME which are commonly based on individual feature contribution techniques), PbCRPE predicts and explains following a similar approach.

As mentioned previously, credit risk prediction depends on multiple features rather than individual features. For instance, a 'high debt-to-income ratio' may only indicate default when combined with 'low job stability'. This observation highlights the need for explanation approaches that consider multiple features. In this context, PbCRPE addresses this need by using contrast patterns to explain predictions. Contrast patterns allow providing explanations based on conjunctions of *feature#value* conditions, similar to those used by decision trees. This structural similarity motivates the use of a predictor based on decision trees, not only because of their alignment with contrast patterns, but also because of their strong performance in state-of-the-art credit risk prediction [38]. A decision tree can be interpreted as a hierarchy of decision rules, where each path from the root to a leaf node corresponds to a conjunction of conditions of the form *feature#value*. Similarly, contrast patterns are expressed as conjunctions of conditions of the form *feature#value*. Based on this relationship, the pro-

posed algorithm is organized into two main stages: (i) a training stage, which includes predictor training and contrast pattern mining from the classes, and (ii) a prediction and explanation stage using the trained predictor and the mined contrast patterns.

### 3.1 Training and contrast pattern mining stage

The first stage of the PbCRPE consists of training a credit risk predictor and mining contrast patterns, which are expressed as multiple *feature#value* conditions that will later be used to explain the classifier's predictions.

For credit risk prediction, we propose using the FT4cip predictor [55], a decision tree-based predictor capable of handling mixed and imbalanced datasets, which has reported strong predictive quality on such data, making it suitable for this task. Although FT4cip is designed to handle imbalanced datasets, the experiments conducted in this work indicated that it achieved better predictive quality when trained using a previously balanced training set. Thus, following the results reported in [56], we propose applying SMOTENC [50, 57] to the training set to balance it prior to predictor training.

For mining contrast patterns, since instances belonging to the same class do not necessarily share the same *feature#value* conditions and may form different groups of similar instances, these groups are first identified through clustering. Contrast patterns are subsequently mined independently for each cluster with respect to the whole opposite class, yielding contrast patterns specific to both the cluster and its associated class. To generate these clusters, we propose using the Kmodes algorithm [58, 59], as it is specifically designed for mixed data and avoids the limitations of distance-based methods when handling non-numeric features. To determine the number of clusters, following [60, 61], we propose using the Silhouette coefficient, which, according to these works, enables the identification of well-defined clusters.

For mining contrast patterns, we propose using CPMining, introduced in the PBC4cip algorithm [62, 63], as a supervised contrast pattern miner that explicitly exploits class differences to identify contrast patterns. CPMining is applied independently to each cluster by contrasting the instances of the cluster with all instances of the opposite class. This design allows mining contrast patterns that represent *feature#value* conditions observed among similar instances within each cluster while appearing less in instances from the opposite class.

Due to the large number of contrast patterns that can be generated during mining, the filtering used in CPMining is applied to reduce redundant patterns. This filtering removes more specific patterns when a more general pattern covers the same instances, and when redundant conditions are present within a pattern. Additionally, we propose removing contrast patterns whose support in the op-

posite class is non-zero. As a result, the final set contains only contrast patterns whose support in the opposite class is zero.

### 3.2 Prediction and explanation stage

Given a new instance  $x$ , the prediction and explanation stage first determines its class (defaulter or non-defaulter) using the trained FT4cip predictor. Once the predicted class  $c$  for  $x$  has been determined, the contrast patterns mined from each cluster associated with class  $c$  are used to compute a similarity score between the new instance  $x$  and each cluster. The cluster with the highest similarity score is selected as the cluster most similar to the new instance.

To identify the cluster that best represents a new instance  $x$ , PbCRPE defines a contrast pattern-based similarity score between  $x$  and each cluster  $C_{c,j}$  associated with the predicted class  $c$ . Unlike conventional similarity measures for mixed data, which compare instances directly using their feature values, the proposed score evaluates similarity through the contrast patterns mined from each cluster. These patterns summarize the characteristic combinations of *feature#value* conditions shared by similar instances within the cluster and associated with the predicted class, making them a more appropriate representation for explanation purposes. Consequently, the similarity between  $x$  and a cluster is quantified according to the support accumulated by the cluster’s contrast patterns that cover the new instance. Specifically, the similarity score is computed as the ratio between the accumulated support of the covering contrast patterns and the total support of all contrast patterns mined from the cluster, as defined in Equation 1.

$$Sim(x, C_{c,j}) = \frac{\sum_{p \in \mathcal{P}_{c,j}(x)} supp(p)}{\sum_{p \in \mathcal{P}_{c,j}} supp(p)}, \quad (1)$$

This similarity score increases as the support of the contrast patterns covering the new instance increases. Since contrast patterns with higher support cover more similar instances within the cluster, they contribute more to the similarity score. Furthermore, dividing by the total support of the contrast patterns mined from the cluster avoids bias towards clusters with more contrast patterns. Consequently, higher similarity scores indicate that a larger proportion of the support of the contrast patterns mined from the cluster is associated with to contrast patterns that cover the new instance.

The cluster with the highest similarity score is selected as the cluster most similar to the new instance  $x$ . If multiple clusters obtain the same similarity score, the cluster whose highest-support contrast pattern covers the new instance is selected. This criterion favors the cluster whose highest-support contrast pattern covers the largest number of similar instances. The explanation stage assumes that at least one contrast pattern mined from the clusters associated with the

predicted class covers the new instance. Under this assumption, the most similar cluster always contains at least one covering contrast pattern from which the explanation can be selected. The empirical validity of this assumption is analyzed in Section 4. Once the most similar cluster  $C_{c,j^*}$  has been identified, the contrast pattern  $p^*$  with the highest support is selected from  $\mathcal{P}_{c,j^*}(x)$ , the set of contrast patterns from  $C_{c,j^*}$  that cover the instance  $x$ , as defined in Equation 2. Support is used as the selection criterion because contrast patterns with higher support cover a larger number of similar instances within the selected cluster, which means that more instances share the same *feature#value* conditions.

$$p^* = \arg \max_{p \in \mathcal{P}_{c,j^*}(x)} \text{supp}(p), \quad (2)$$

The selected contrast pattern becomes the final explanation of the prediction and is expressed as a conjunction of *feature#value* conditions. Since the selected contrast pattern covers multiple similar instances within the most similar cluster, the explanation is based on conditions shared by the new instance and those covered instances. If multiple patterns achieve the same maximum support, the pattern containing the largest number of *feature#value* conditions is selected, since it provides a more detailed explanation by incorporating a larger set of conditions associated with the covered instances. If multiple contrast patterns remain tied after applying this criterion, all tied contrast patterns will be treated as the explanation.

## 4 Experimental Results

The objective of this experimental evaluation is to assess whether PbCRPE can achieve competitive prediction quality while simultaneously providing explainable predictions. The evaluation considers two complementary aspects: (i) prediction quality and (ii) explanation capability. Prediction quality is assessed using ROC AUC and Type II Error, two commonly used metrics in credit risk prediction that capture complementary aspects of predictive quality: the overall ability of the predictor to discriminate between classes and its ability to correctly identify defaulters, respectively.

PbCRPE is compared with twelve recent state-of-the-art credit risk prediction methods under a common experimental protocol using twelve public and private credit risk datasets. These methods were selected from studies published between 2024 and 2025 because they represent recent developments in machine-learning-based and explainable approaches to credit risk prediction, report competitive predictive results, and evaluate their approaches on benchmark datasets that overlap with or are comparable to those considered in this work. Consequently, the comparison provides a representative and up-to-date evaluation of PbCRPE against recent approaches reported in the literature.

The explanation capability of PbCRPE is evaluated by analyzing the contrast patterns generated for individual predictions. In particular, the analysis examines whether the selected contrast patterns provide specific combinations of *feature#value* conditions associated with the predicted class and shared between the new instance and similar instances within the selected cluster.

#### 4.1 Datasets

The experimental evaluation was conducted using twelve credit risk datasets obtained from both public and private sources. The datasets were selected because they represent diverse credit risk scenarios while sharing the two characteristics addressed by the proposed PbCRPE algorithm: mixed-type features (numerical and categorical) and different degrees of imbalance. The diversity in attribute types and class distributions allows the behavior of PbCRPE to be evaluated under different data conditions.

Table 1 summarizes the main characteristics of the datasets, including the number of instances, number of features, features types, class distribution, imbalance ratio (IR), and data source. Most of the public datasets were obtained from the UCI Machine Learning Repository<sup>1</sup> and Kaggle<sup>2</sup>, two widely used repositories for benchmarking credit risk prediction methods. The datasets vary substantially in both size and class distribution, ranging from 628 to 150,000 instances and from nearly balanced datasets (IR = 1.09) to highly imbalanced datasets (IR = 13.96). Consequently, the experimental evaluation covers a broad spectrum of credit risk scenarios, allowing the prediction quality of PbCRPE to be assessed under different levels of imbalance.

Table 1: Characteristics of the twelve credit risk datasets used in the experimental evaluation.

| Dataset        | Instances | Features | Numerical/Categorical | No-defaulter | Defaulter | IR    | Domain  |
|----------------|-----------|----------|-----------------------|--------------|-----------|-------|---------|
| AER            | 1,319     | 12       | 9/3                   | 1,023        | 296       | 3.45  | Public  |
| Australia      | 690       | 14       | 6/8                   | 307          | 383       | 1.24  | Public  |
| Japan          | 690       | 15       | 6/9                   | 307          | 383       | 1.24  | Public  |
| EIZ            | 5,510     | 22       | 13/9                  | 2,295        | 3,215     | 1.40  | Private |
| German         | 1,000     | 20       | 7/13                  | 700          | 300       | 2.33  | Public  |
| HMEQ-Data      | 3,364     | 12       | 10/2                  | 3,064        | 300       | 10.21 | Public  |
| Taiwan         | 30,000    | 24       | 20/3                  | 23,364       | 6,636     | 3.52  | Public  |
| GMSC           | 150,000   | 11       | 10/1                  | 139,974      | 10,026    | 13.96 | Public  |
| Heloc          | 10,459    | 23       | 23/0                  | 5,000        | 5,459     | 1.09  | Public  |
| Loan data set  | 628       | 12       | 7/4                   | 332          | 296       | 1.12  | Public  |
| PPDai          | 55,596    | 29       | 22/7                  | 48,413       | 7,183     | 6.74  | Public  |
| Thomas dataset | 1,225     | 14       | 14/0                  | 902          | 323       | 2.79  | Public  |

<sup>1</sup> <https://archive.ics.uci.edu/>

<sup>2</sup> <https://www.kaggle.com/>

## 4.2 Prediction Quality

Figures 1 and 2 compare the prediction quality of PbCRPE with that of the twelve selected state-of-the-art methods using ROC AUC and Type II Error, respectively. Figure 1 reports the mean ROC AUC across the twelve evaluated datasets, whereas Figure 2 presents the corresponding mean Type II Error across the same datasets.

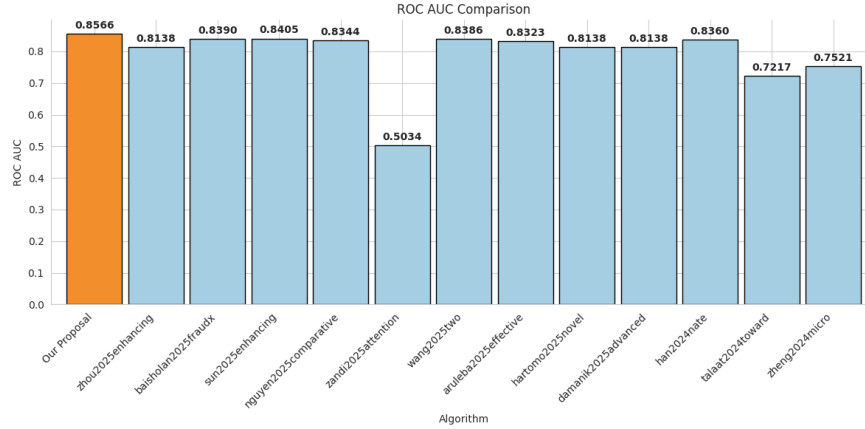


Fig. 1: Mean ROC AUC obtained by PbCRPE and the compared state-of-the-art methods across the twelve evaluated credit risk datasets. Higher values indicate better prediction quality.

Across the twelve evaluated datasets, PbCRPE achieves a mean ROC AUC of **0.8566** and a mean Type II Error of **0.1432**. These results indicate that the proposed algorithm \*\*provides competitive prediction quality compared with\*\* the selected state-of-the-art methods. In particular, PbCRPE attains the highest ROC AUC on several datasets while maintaining competitive results on the remaining ones. Similarly, PbCRPE achieves a lower Type II Error than competing methods on multiple datasets, indicating a lower rate of actual defaulters being incorrectly classified as non-defaulters.

The differences observed between ROC AUC and Type II Error are expected because both metrics evaluate complementary aspects of prediction quality. ROC AUC measures the overall ability of the predictor to discriminate between defaulters and non-defaulters across different decision thresholds, whereas Type II Error specifically measures the proportion of actual defaulters incorrectly classified as non-defaulters.

From a practical perspective, minimizing Type II Error is particularly important in credit risk prediction because failing to identify a defaulter may result

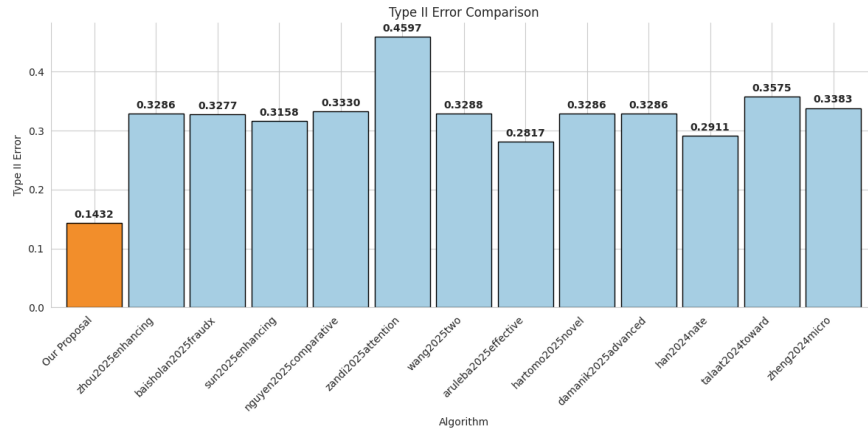


Fig. 2: Mean Type II Error obtained by PbCRPE and the compared state-of-the-art methods across the twelve evaluated credit risk datasets. Lower values indicate better prediction quality.

in direct financial losses for lending institutions. Therefore, the Type II Error obtained by PbCRPE reflects its ability to reduce the proportion of defaulters that are incorrectly classified as non-defaulters, while maintaining competitive overall discrimination capability.

### 4.3 Explanation Analysis

The explanation analysis evaluates whether the contrast pattern mining algorithm provides sufficient pattern coverage for the instances evaluated by PbCRPE. In particular, we verify whether, for each evaluated instance  $x$ , at least one contrast pattern mined from a cluster associated with the predicted class covers  $x$ . This analysis directly evaluates the coverage assumption defined for the prediction and explanation stages. A prediction is considered covered when  $\mathcal{P} * c, j(x) \neq \emptyset$  for at least one cluster  $C * c, j$  associated with the predicted class  $c$ .

Across the twelve evaluated credit risk datasets, all evaluated instances were covered by at least one contrast pattern mined from a cluster associated with the predicted class. Consequently, the explanation stage successfully identified a covering contrast pattern for every evaluated prediction, and no fallback mechanism was required. This result provides empirical evidence that the proposed \*\*clustering and contrast pattern mining approach provides complete pattern coverage across\*\* the evaluated datasets.

Although this observation does not constitute a theoretical guarantee that every possible future instance will be covered, it confirms the empirical validity

of the coverage assumption under the experimental conditions considered in this study.

#### 4.4 Discussion

PbCRPE achieves competitive prediction quality compared with the twelve evaluated state-of-the-art approaches while simultaneously providing an explanation for each prediction. This dual capability, integrating prediction and explanation within a single algorithm, represents an important advantage over post-hoc explanation techniques such as SHAP and LIME, which typically express explanations through feature-level importance scores rather than explicitly representing the joint effect of multiple feature-value conditions. This distinction is particularly relevant in credit risk prediction, where default risk is generally associated with combinations of financial and personal attributes rather than individual features considered in isolation\*\* [38].

To illustrate the explanation process, Table 2 presents an example of an instance from the German credit dataset that PbCRPE classified as *defaulter credit risk*.

Table 2: Example of a new instance predicted as bad credit risk.

| Attribute       | Value               |
|-----------------|---------------------|
| Checking status | No checking account |
| Duration        | 36 months           |
| Credit history  | Critical account    |
| Purpose         | Radio / Television  |
| Credit amount   | \$3,342             |
| Interest rate   | 4%                  |
| Existing loans  | 1                   |

For the instance shown in Table 2, PbCRPE first considers the clusters associated with the predicted class of the instance. Using the contrast pattern-based similarity score defined in Equation 1, the cluster most similar to the new instance is selected. The following contrast pattern is then selected from among the patterns in this cluster that cover the new instance, according to the highest-support criterion defined in Equation 2:

$$\begin{aligned} & \text{credit\_history} \neq \text{A34} \wedge \text{amount} < \$7,824 \wedge \text{duration} > 28 \text{ months} \\ & \wedge \text{interest\_rate} > 2\% \wedge \text{num\_loans} \leq 1 \end{aligned}$$

Thus, the selected contrast pattern covers the instance because all of its *feature#value* conditions are simultaneously satisfied. For example, the condition *credit\_history*  $\neq$  A34 is satisfied because the instance has a *Critical account* credit history, which corresponds to a value different from A34. Similarly, *amount*  $<$  \$7,824 is satisfied because the credit amount is \$3,342; *duration*  $>$  28, months is satisfied because the duration is 36 months; *interest\_rate*  $>$  2%

is satisfied because the interest rate is 4%; and  $num\_loans \leq 1$  is satisfied because the instance has one existing loan. Since a contrast pattern is defined as a conjunction of conditions, all these conditions must be satisfied simultaneously for the pattern to cover the instance. Importantly, the pattern is not interpreted as a collection of independent feature contributions. Rather, its conjunction represents a combination of *feature#value* conditions shared by the new instance and multiple similar instances within the selected cluster. Consequently, the contrast pattern provides a concise representation of the combination of conditions associated with the predicted credit risk class.

The example also illustrates the role of performing contrast pattern mining at the cluster level rather than directly at the class level. Instead of extracting patterns intended to represent all instances belonging to the same class, PbCRPE mines contrast patterns from groups of similar instances. This strategy allows different local structures within the same class to be captured, enabling the resulting patterns to represent *feature#value* conditions that are more specific to particular groups of similar instances.

The example also illustrates how the two-step selection strategy contributes to the resulting explanation. First, the predicted class determines the clusters considered for the new instance, and the similarity score identifies the cluster whose covering contrast patterns provide the greatest similarity to the instance. Subsequently, among the contrast patterns from the selected cluster that cover the instance, the pattern with the highest support is selected as the final explanation. Thus, the explanation is first associated with the group of similar instances that is most similar to the new instance and is then determined by the most supported combination of *feature#value* conditions within that group.

The experimental results confirm that the coverage condition required by the explanation procedure was satisfied for all evaluated instances. In particular, for every instance, at least one contrast pattern mined from a cluster associated with the predicted class covered the instance. Therefore, the similarity score could be computed based on covering contrast patterns for all evaluated instances, and the explanation procedure was successfully applied without requiring any fallback mechanism. These results provide empirical evidence that the coverage condition defined in the explanation procedure was consistently satisfied under the experimental conditions considered in this study.

Taken together, these results indicate that PbCRPE combines competitive prediction quality with explanations based on contrast patterns that represent combinations of *feature#value* conditions shared by similar instances. By integrating prediction for mixed-type and imbalanced data with cluster-level contrast pattern mining and representative pattern selection, PbCRPE generates explanations that are directly associated with groups of similar instances rather than isolated feature contributions. These characteristics make the proposed al-

gorithm suitable for credit risk datasets involving mixed-type features and class imbalance, where both prediction quality and explainable decision support are relevant.

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